



# Maxime Dherbecourt

PhD Student in Physics

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## PROFILE

I'm a PhD student based at the IPCMS laboratory in Strasbourg (France), working on strong light-matter coupling and its effects on physical phenomena such as Raman scattering and transport properties. I enjoy making science accessible through interactive explorations and visual explanations.

**Research interests:** Light-Matter Interactions · Condensed Matter · Open Quantum Systems · Numerical Methods

## EDUCATION

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| <b>PhD in Physics – Strong Light-Matter Coupling</b><br>Institut de Physique et Chimie des Matériaux de Strasbourg (IPCMS) · France<br>Advisors: Guillaume Weick and David Hagenmüller. | 2024 – Now  |
| <b>Master Degree – International Center for Fundamental Physics (ICFP)</b><br>Ecole Normale Supérieure de Paris · France<br>Condensed Matter track. Graduated with honors.              | 2023 – 2024 |
| <b>Magistère de Physique Fondamentale</b><br>Université de Strasbourg · France                                                                                                          | 2021 – 2024 |
| <b>Bachelor Degree – Advanced Mathematics and Physics</b><br>Université de Strasbourg · France                                                                                          | 2019 – 2022 |

## TEACHING

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|--------------------------------------------------------------------------------------------------|-------------|
| <b>Electromagnetism 1</b><br>Lecturer · Université de Strasbourg – General Science License       | 2024 – 2026 |
| <b>French Physicists' Tournament</b><br>Assistant · Université de Strasbourg – Master of Physics | 2024 – 2026 |

## INITIATIVES

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|-----------------------------------------------------------------------------------------------------------------------------|------------|
| <b>French Physicists' Tournament</b><br>Organizer · Young Researcher Network – Société Française de Physique                | 2025 – Now |
| <b>Winner of a science popularization contest</b><br>Prize Winner · Summer of Math Exposition – "The Science of Complexity" | 2025       |

## Scientific Game Jam on my PhD project

2025

Participant · Strasbourg – Created game "Molec' Adventure"

## PRESENTATIONS

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### Raman Scattering in the Vibrational Strong Coupling Regime

2026

Poster · Summer School – Open Quantum Systems, Les Houches, France

### Raman Scattering in the Vibrational Strong Coupling Regime

2026

Poster · QMat Days, Strasbourg, France

### Spontaneous Raman scattering under Vibrational Strong Coupling

2026

Contributed Talk · European Materials Research Society Conference, Strasbourg, France

### Spontaneous Raman scattering under Vibrational Strong Coupling

2026

Invited Talk · Namasute Workshop, Odense, Denmark

### Spontaneous Raman scattering under Vibrational Strong Coupling : The critical role of polariton spatial mode coherence

2025

Invited Talk ·  $\hbar$ -Meeting, Strasbourg, France

## PUBLICATIONS

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### *Spontaneous Raman Scattering under Vibrational Strong Coupling: The Critical Role of Polariton Spatial Mode Coherence*

2026

M. **Dherbécourt**, J. Bellessa, C. Symonds, G. Weick and D. Hagenmüller

arXiv preprint, arXiv:2511.01723 · [doi.org/10.48550/arXiv.2511.01723](https://doi.org/10.48550/arXiv.2511.01723)

### *Symmetry-controlled ultrastrong phonon–photon coupling in a terahertz cavity*

2026

D. Kim, M. **Dherbécourt**, et al.

The Journal of Chemical Physics, 164, 104201 · [doi.org/10.1063/5.0313120](https://doi.org/10.1063/5.0313120)

## PERSONAL PROJECTS

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### WaterDropEngine Terrain Generator

2026

A procedural terrain generator built with WaterDropEngine, producing realistic 3D landscapes using noise-based algorithms and GPU-accelerated mesh generation.

Rust · Procedural Generation · Terrain · Vulkan · [github.com/explorablescience/WaterDropTerrainGenerator](https://github.com/explorablescience/WaterDropTerrainGenerator)

### Portfolio Website

2025

The source code for this portfolio website, built with Next.js, TypeScript, Tailwind CSS and Vercel. It features a clean and modern design, showcasing my projects and skills.

Next.js · TypeScript · Tailwind CSS · Vercel · React · [github.com/xam4lor/Website](https://github.com/xam4lor/Website)

### WaterDropEngine

2023

A high-performance, ECS driven and Vulkan based game engine written in Rust. Designed for flexibility and speed, with a focus on 3D game development.

Rust · Game Engine · Vulkan · WebGPU · [github.com/explorablescience/WaterDropEngine](https://github.com/explorablescience/WaterDropEngine)

### Explorable Science

2019

Explorable visual articles and interactive simulations on various scientific topics such as phase transitions, general relativity, quantum mechanics and more.

Science Communication · Interactive Simulations · Explorables · [explorablescience.com](https://explorablescience.com)

## Physics Simulation Engine

2017

A high-level JavaScript library for creating interactive physics simulations in the browser built on top of the canvas API.

JavaScript · Physics Simulation · Educational · Interactive Science · [github.com/explorablescience/PhysicsSimulationEngine](https://github.com/explorablescience/PhysicsSimulationEngine)

## SKILLS

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**Programming Languages:** Julia, Rust, JavaScript/TypeScript, HTML5/CSS3, C++, Python3, Java

**Programming Libraries & Frameworks:** Vulkan / WebGPU / OpenGL, PyTorch / NumPy, Node.js / React.js / Three.js / React Three Fiber / Next.js, Plotly.js / p5.js / Processing, CUDA

**Scientific tools:** Mathematica, Maple

**Other:** LaTeX, Git, Docker

## LANGUAGES

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**French:** Native · **English:** Fluent (C1) · **German:** Intermediate (B1)